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A Comparative Analysis of Pre-trained Models for Identifying Alzheimer's Disease

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***Abstract*—Neurodegenerative diseases like Alzheimer's disease (AD) remain a critical global healthcare challenge, highlighting the importance of early and accurate detection. Deep learning models, especially transfer learning (TL), leverage pre-trained weights, enhancing model performance with limited data. These models are trained on various images, and each task requires fine-tuning. This study evaluates three prominent pre-trained architectures, ResNet50, EfficientNetB0, and InceptionV3, to compare their precision and accuracy in image classification. These three models are fine-tuned to diagnose AD efficiently. The four-class unbalanced dataset used for our task is collected from the Open Access Series of Imaging Studies-1 (OASIS-1). Synthetic Minority Oversampling Technique (SMOTE) is performed to convert the dataset into a balanced one. This dataset was then fed to the fine-tuned models, and accuracy was calculated for each model. The results were also compared with one another and it was found that EfficientNetB0 outperformed the others in all aspects, having a validation accuracy and loss of 99.65% and 1.18%, respectively. Its performance was found superior to some of the state-of-the-art models also tried by other prominent researchers.**

***Index Terms*—Alzheimer's disease, OASIS-1, MRI, SMOTE, Transfer learning, Deep learning**

I. Introduction

Managing healthcare effectively remains one of the most significant challenges across the globe, largely due to various global factors, including the growing aging population. As life expectancy continues to rise, elderly-associated disorders such as dementia as well as AD are becoming increasingly widespread. Due to this increasing number of affected aged persons, healthcare systems and caregivers are facing immense pressure day by day. AD, a neurodegenerative ailment, leads to cognitive decay, memory deprivation, and reduced decision-making abilities, ultimately diminishing an individual's quality of life. A new case of dementia is identified somewhere in the world every three seconds. In 2020, the condition impacted over 55 million people around the world and it is expected that this number will reach approximately double by 2030 [1].

AD leads to neuron degeneration in the brain's cortical region. Two key proteins, beta-amyloid, and tau, are responsible for clearing cellular debris within neurons. However, in individuals affected by AD, these proteins become toxic to brain function. As time progresses, tau proteins form abnormal tangles inside neurons, and beta-amyloid forms plaques between nerve cells. Consequently, neurons struggle to communicate efficiently, ultimately resulting in cell death. This process particularly affects the hippocampus, the brain's center for learning and memory retention, causing it to shrink progressively. The ventricles expand in the affected brain, while the cerebral cortex and hippocampus shrink. MMSE (The Mini-Mental State Examination) and CDR (Clinical Dementia Rating) are widely utilized to implement a consistent approach to grading dementia severity [2].

As no cure exists for AD, identifying it initially is the only way to manage the condition and potentially slow its progression. The later stages of Alzheimer's disease often necessitate prolonged hospitalization and extensive ongoing care, which are highly costly. Detecting the disease in its early stages can help mitigate these expenses. The application of DL (deep learning) and ML (machine learning) in AD diagnosis has gained attention recently. Data used for these model training and detection includes mammography, EEG, microscopy, and Magnetic Resonance Imaging (MRI) [3]. As MRI can easily detect the shrinking of brain cells, along with being safe, painless, and delivering high-quality images of neural tissues.

Building a new model is a tough task as it needs to be trained rigorously on the dataset, and a large training dataset is required. However, data scarcity is a common challenge when working with MRI images as this is a costly as well as expertise-oriented task. TL is the way out of this problem as it minimizes the need for extensive training data. A pre-trained model, having already been trained on vast amounts of data, and fine-tuned the model to make it compatible enough to help our cause. It reduces our need for training on massive amounts of data to ensure optimal accuracy in the automation of AD detection.

This research is motivated by the critical need to improve the prompt detection of AD using Sophisticated DL models. By leveraging neuroimaging information, the goal is to improve diagnostic accuracy, enabling better treatment options and an elevated quality of life for those impacted. In this survey, we have tried three different TL models with fine-tuned layers to make AD detection easy and less complex. Multi-class MRI data was utilized for this task, with the classification categories including Non Demented (ND), Very Mild Dementia (VmD), Mild Dementia (MD), and Moderate Dementia (MoD). ResNet50, EfficientNetB0, and InceptionV3 models were used in this case to detect AD. Results show that all three models perform significantly well after fine-tuning compared to other earlier works in the same area. The succeeding sections outline the organization of this survey: Section II offers a study of related works. Section III outlines the dataset. Section IV highlights the methodology, detailing the image pre-processing, classification techniques for AD, the use of TL methods for disease detection, and model fine-tuning. Section V shows the results and discussions, including the research design and experimental findings. Finally, Section VI concludes the study by summarizing key contributions and proposing directions for forthcoming research.

II. Related Work

DL techniques and CNN models have revolutionized the identification of AD using brain MRI images. Many researchers are exploring and developing different CNN models to detect early signs of AD, paving the way for preventive care. Sisodia et al. [4] survey paper explored the different pre-trained models like ResNet50, VGG19, Xception, DenseNet201, and EfficientNetB7 using an MRI multi-class AD dataset. After some pre-processing and data augmentation techniques, the DenseNet201 model outperformed others with an impressive validation performance of 96.59%. Yadav et al. [5] used advanced technology and attention mechanisms with CNN to improve the detection accuracy. Their research emphasizes advanced identification of AD and CNN along with the attention mechanism, making their model accuracy 95% to 98.54%. Suriya et al. [6] proposed DEMNET, a CNN model to identify AD, where two 2D convolution layers and four DEMNET blocks are used for feature extraction. Datasets are collected from ADNI, and SMOTE is used for balancing the datasets. Both DEMNET and attention mechanisms enhance feature extraction, but DEMNET's dense layers increase computational load and training time, while attention suffers from high memory usage, computational cost, and lacks inductive biases.

Shamrat et al. [7] comprehensive survey presented AlzheimerNet, a fine-tuned CNN model that diagnoses MRI images, acquired from the ADNI dataset. For better performance, the CLAHE image enhancement method is applied, and after that, an augmentation process is performed to overcome the data unbalanced problem. Rao et al. [8] presented a Multi-Layer Perceptron along with SVM for feature extraction and classification of AD. This research utilizes coronal, axial, and sagittal 2D orientations from a 3D MRI dataset. Tian et al. [9] highlighted the generative adversarial network-driven brain section image improvement and CNN predictive algorithm for detecting AD in the primary stage. The study was done for ADNI and OASIS datasets. The brain slice GAN model for the affected brain detection approach extracts high-level brain features to diagnose this disease.

AD can be detected from EEG signals, which is another comprehensive method in the diagnosis field. Jiao et al. [10] deployed a machine learning model and EGG for early detection of AD. Researchers extracted from resting-state EEG recordings and applied Random Forest Regression to identify three label classifications. EEG biomarkers, CSF biomarkers, and APOE phenotype are explored for accurate results, and EEG outperformed others. Sankari et al. [11], investigated wavelet coherence EEG to differentiate a healthy brain from an AD-affected brain. In this research, Wavelet coherence of the electrode is computed across various frequency bands (delta, theta, alpha, and beta), and significant features are found for AD-affected brains. According to Bagaskara et al., [12], the initial stage of AD is the most important stage to prevent this disease. They research pre-trained model VGG-16 and VGG-19 architectures for early detection of different stages of AD. Their survey focused on real-time cloud-based CNN and achieved 99.38% validation accuracy. However, these approaches often suffer from limitations such as inadequate handling of class imbalance, suboptimal hyperparameter selection, and a lack of statistically validated comparisons. In contrast, our study presents a fine-tuned TL approach using optimized hyperparameters (learning rate, batch size, activation functions, and optimizer choice), and this sets our work apart as a systematically validated, high-performing, and practically applicable solution for multi-class AD stage classification.

III. Dataset

We utilized the OASIS-1 [13] dataset in our survey, which provides brain imaging and clinical information for research on AD and other neurodegenerative conditions. The dataset is categorized into four labels based on CDR values: ND, VmD, MD, and MoD. The images were acquired from 416 subjects using a 1.5T MRI scanner, and all these details can be found in Table I.

TABLE I
Dataset details

Label	CDR Score	Subjects	Age Range	Total MRI
ND	0.0	228	18-96	67,222
VmD	0.5%	100	60-96	13,725
MD	1.0%	66	60-96	5,002
MoD	2.0%	22	60-96	488

Firstly, an MRI produces a 3D volumetric image of the brain. Then this image is divided into multiple 2D slices for analysis, in three anatomical planes, axial, coronal, and sagittal. In this dataset, the brain images were sliced into 256 slices according to the axial plane (top to bottom). These T1-weighted images provide high-resolution details of the brain's anatomical structure, which are crucial for studying brain atrophy (shrinkage).

IV. Methodology

A Multilayer Perceptron consists of a fully linked, forward-propagation neural network where neurons in one level are linked to every neuron in the subsequent level. However, this design has drawbacks, such as a large number of Hyperparameters, high computational cost, and loss of spatial information. Convolutional Neural Networks (CNNs) effectively address the shortcomings of MLPs by processing inputs as tensors rather than flattened vectors. This allows CNNs to reduce the number of parameters, preserve spatial information, and leverage local feature detection.

A. Transfer Learning

CNNs are trained using the process of the backpropagation algorithm, where each neuron updates its weight after every iteration to minimize the computational cost. However, if the dataset is low, the CNN model's efficiency decreases because of overfitting or underfitting. In this scenario, TL is a more effective strategy, utilizing a model which is initially learned on a vast data sample like 'ImageNet'. In the early layers, general patterns (edges, textures, shapes), are retained and the later layers are modified or retrained on a smaller, task-specific dataset.

Feature extraction involves using a pre-trained network to obtain useful features without changing the pre-trained weights from the input data. All layers except the final fully connected layer are "frozen," meaning their weights are not updated during training, and the final layer is set according to the task output. Fine-tune involves unfreezing some of the layers and allowing them to adapt to the new dataset by updating their weights during training. This process is more appropriate when researchers have insufficient data for the new task and want the model to learn task-specific features from the pre-trained knowledge. TL enables us to leverage well-optimized network architectures, reducing training time, improving generalization, and enhancing performance even with limited data. A schematic of our recommended methodology is shown in Fig. 1

1) ResNet50 Model: ResNet-50 is a 50-layer deep CNN network, well-known for its ability to develop very deep architectures while avoiding the vanishing gradient problem. Kaiming et al. [14] introduced Residual Networks (ResNet) by Microsoft. ResNet introduces skip connections, enabling gradients to propagate directly through the network, enabling deeper architectures. ResNet-50 composed of 49 convolution layers, 16 residual blocks, and a single fully connected layer.

2) EfficientNetB0 Model: EfficientNetB0 is part of the EfficientNet family, introduced by Google AI [15] in 2019. It is a lightweight yet powerful CNN that achieves high accuracy with fewer parameters compared to other traditional models. EfficientNetB0 is designed using 18 convolution layers, 16 Mobile Inverted Bottleneck Convolution blocks, 16 Squeeze-and-Excitation layers, and 1 fully connected layer, making it highly efficient while maintaining accuracy.

3) Inception V3 Model: Inception v3 is a 48-layer CNN developed by Google [16] as part of the Inception family (also called GoogLeNet). Inception V3 consists of 42 convolution layers, 3 types of inception layers with 10 blocks, and 1 fully connected layer. Instead of using large convolutional kernels, Inception v3 breaks them into smaller factorized convolutions.

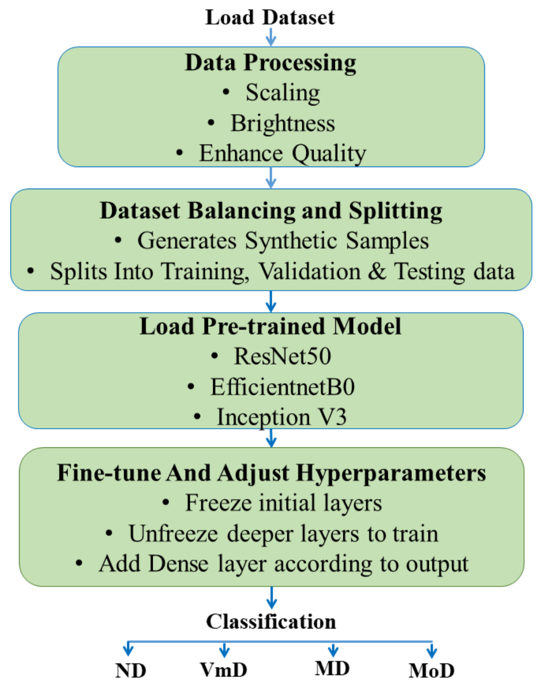

Fig. 1. Visual representation of the research work

This model helps train dense layers by addressing vanishing gradient issues as well as an efficient feature extraction without losing too much information.

We evaluated these 3 state-of-the-art CNN models as TL in our survey. All of these models with some tuning have a great impact on classifying images precisely. We used these models in our multi-class AD detection. Model details can be found in Table II.

TABLE II
MODEL FEATURES AND SPECIFICATIONS

Parameters	ResNet50	EfficientNetB0	Inception V3
Hyperparameters	25.6M	5.3M	23.9M
Image Shape	(224, 224, 3)	(224, 224, 3)	(299, 299, 3)
Architecture	Residual Network	Compound Scaling	Inception Modules
Top-5 Accuracy (ImageNet)	93.0%	93.3%	93.7%

When applying TL to a new dataset, two primary methods are commonly used: feature extraction and fine-tuning. In this research, we used the second one, and in this procedure, we unfroze some of the deep layers (important features) of the pre-trained model and allowed them to be updated along with the new layers. The final layer of neurons is set to four, as our output has four labels. In our training, we set a lower learning rate of 0.00001 in overwriting these pre-trained weights slowly so that it does not affect the trained feature. The 'softmax' activation function and the 'Adam' optimizer were employed in multi-class classification. To avoid overfitting, we use 20% dropout in our models.

B. Data Pre-processing and SMOTE

The dataset we used in our experiments was gathered from OASIS-1. Resizing the images to a consistent size was one of the first steps in pre-processing. Images were acquired at 496×248 pixels but most deep learning models like ResNet50, InceptionV3, and EfficientNetB0 require a fixed input size (e.g., 224x224 or 299x299). So, we resized the image as required for different models. The images were normalized and rescaled, converting pixel values to a certain range. We took 3000 images from all classes, but in MoD, the images were insufficient in quantity. As we had a low number of images in this class, we performed SMOTE to balance the dataset. An unbalanced dataset decreases the model's performance and shows lower accuracy. SMOTE generates synthetic samples of the minority class, helping the model learn from all classes more equally. Also, it helps improve generalization and reduces model bias toward majority classes. Then, the balanced dataset was partitioned into an 85:15 proportion for training and validation. Then this training set was fed to the models.

C. Hardware Setup

To efficiently train a neural network, it is essential to have an adequate hardware setup. We used Windows 11 Pro, OS build 22631.4317 operating system based CPU with Intel 12th-generation Core i5-1235U processor. We ran our program in Jupyter Notebook from Anaconda Navigator, for interactive data analysis and ML. Programs were built and run in a TensorFlow v2.10.0 environment, which also has Keras integrated as part of the TensorFlow library.

V. Results and Discussion

This research primarily aims to diagnose AD by applying various TL models, featuring InceptionV3, ResNet50, and EfficientNetB0. Performance assessment in the study is conducted using evaluation measures like precision, recall, and F1 score. In our study, a trio of previously trained models were fine-tuned, and the performances were observed on the same dataset. We performed SMOTE to balance the class distribution, improving the performance in classification tasks. To enhance the model's proficiency, fine-tuning was applied by adjusting the number of layers in each model. This process involves modifying and optimizing specific layers of the pre-trained models to better suit the AD classification task, allowing for improved feature extraction and more accurate predictions.

The accuracy and loss plots of ResNet50 are presented in Fig. 2, and 3. Similarly, Fig. 4, and 5 illustrate the Loss and accuracy visualization for EfficientNetB0, while Fig. 6, and 7 depict the corresponding curves for InceptionV3. We observed that ResNet50, EfficientNetB0, and Inception V3 models achieved a validation accuracy of 99.00%, 99.65%, and 94.69%, respectively. Their corresponding validation losses were 5.11%, 1.18%, and 17.36%, respectively. The values clearly show that EfficientNetB0 surpasses the rest two in both accuracy and loss. Other metrics like precision, F1 score, and recall values were also satisfactory. In terms of hyper-parameters, EfficientNetB0 is a lightweight model, requiring only 898,356 parameters. This improves cost-efficiency by optimizing computational resources and shortening training time. Comparative analysis of these models' performances is

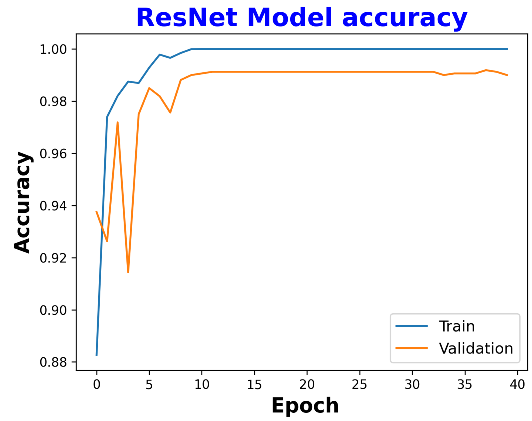

Fig. 2. ResNet50 accuracy plot

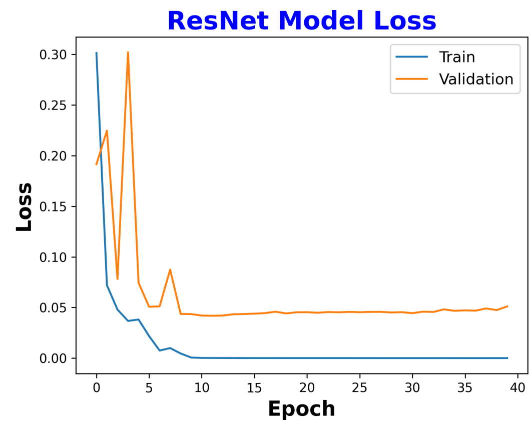

Fig. 3. ResNet50 loss plot

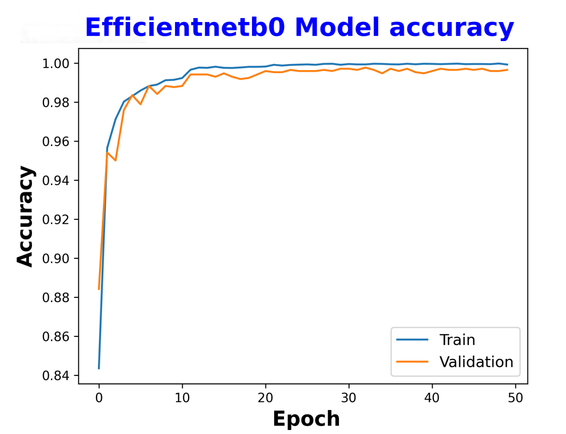

Fig. 4. EfficientNetB0 accuracy plot

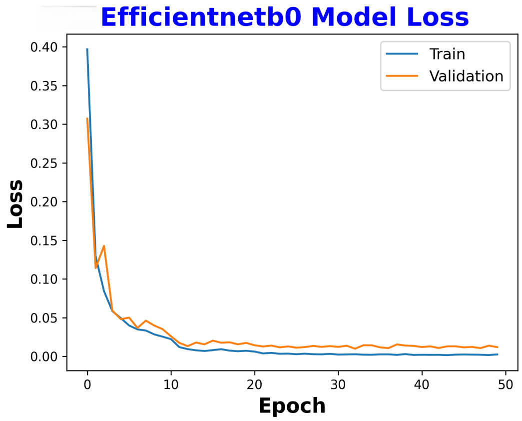

Fig. 5. EfficientNetB0 loss plot

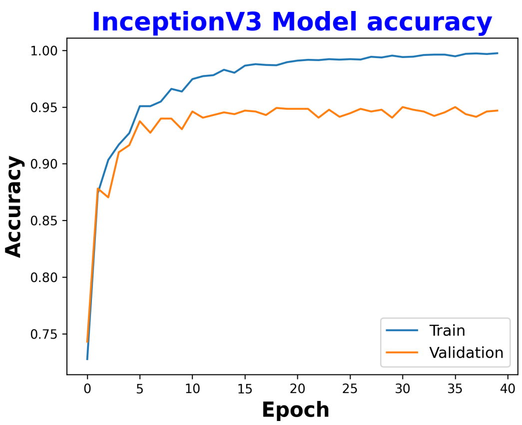

Fig. 6. InceptionV3 accuracy plot

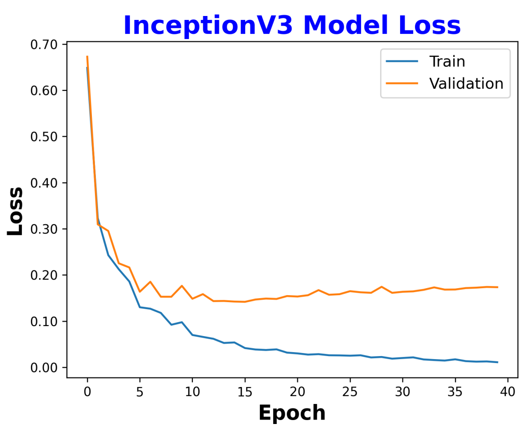

Fig. 7. InceptionV3 loss plot

shown in Table III.

A confusion matrix (CM) furnishes an extensive overview of model outcomes by comparing actual and predicted categories, which is a performance evaluation tool used to assess the accuracy of classification models. Fig. 8 illustrates the CM for

TABLE III
PERFORMANCE EVALUATION AMONG FINE-TUNED MODELS

Model	ResNet50	EfficientNetB0	Inception V3
Accuracy	99.0%	99.65%	94.69%
Loss	5.11%	1.18%	17.36%
Precision	99.33%	99.41%	93.11%
F1 Score	98.67%	99.02%	95.15%
Recall	98.16%	99.28%	94.56%
Parameters	4,473,860	898,356	2,198,141

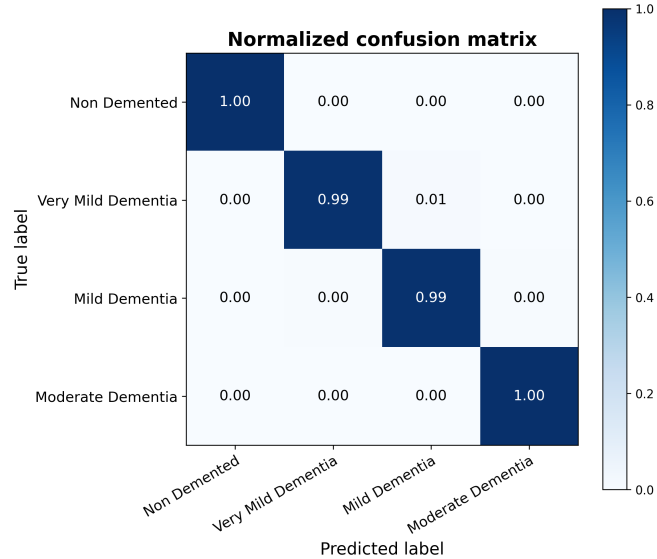

Fig. 8. Confusion Matrix of EfficientNetB0

EfficientNetB0, which shows high values along the diagonal (where actual and predicted labels match) and low values in the off-diagonal elements (where misclassifications occur). This indicates that the model is making accurate predictions with minimal errors. Similarly, we have compared our study with some other works done recently by researchers to prove the novelty of our work. The comparison can be seen in Table IV. From the data, it is evident that our fine-tuned TL model overcame the data imbalance problem and worked better than other TL models developed on the same data sample. The proposed model was fine-tuned using the Adam optimizer with a learning rate of 0.00001 and SoftMax activation for the output layer. A batch size of 90 was used, which was empirically found to yield the best performance in terms of accuracy and convergence stability in our study.

Notably, the model demonstrated improved sensitivity in detecting early-stage AD, which is clinically significant for timely diagnosis and intervention. However, we only compared three TL models: InceptionV3, ResNet50, and EfficientNetB0. Although these are popular and effective models, testing more advanced or newer models, like Vision Transformers, might give better results or new insights. Finally, using techniques

TABLE IV
ASSESSMENT OF THE PROPOSED FRAMEWORK AND STATE-OF-THE-ART MODELS

Reference	Dataset	TL Model	Accuracy
Ghosh et al. [17]	OASIS	MobileNet	95.24%
Fulton et al. [18]	OASIS	ResNet50	98.99%
Ruchi et al. [19]	OASIS	VGG16	94.4%
Marcia et al. [20]	OASIS	VGG16, Inceotion V4	92.3% 96.25%
Bijen et al. [21]	OASIS	AlexNet, GoogleNet, ResNet50	94.64% 88.99% 94.35%
Proposed Fine-tuned Transfer Learning Models	OASIS	ResNet50, EfficientnetB0, Inception V3	99.00% 99.65% 94.69%

like attention maps can help explain how the model makes decisions, which is important for supporting real-world use in clinics.

VI. CONCLUSION

This paper presents significant research to identify the most effective transfer learning models for AD identification. Our research presents a performance comparison of three fine-tuned transfer learning models along with other models and methods developed by researchers. Results show that our fine-tuned EfficientNetB0 performed well enough to detect AD returning the highest validation accuracy of 99.65%. This model also has the least number of hyperparameters, which makes it lightweight and easy to apply in a moderately computational powered system. This also saves time and computational cost, another point to consider when applying any model in real-life applications. It is evident from the discussions that fine-tuned transfer learning models can significantly reduce our training load and contribute to building an efficient model. In the future, we plan to explore Vision Transformer (ViT) as a transfer learning approach and evaluate its performance on our dataset. Additionally, we aim to experiment with ensemble models, combining multiple architectures to enhance classification accuracy and overall model robustness.

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